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DEPARTMENT OF ELECTRICAL ELECTRONIC AND COMMUNICATION ENGINEERING

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PROJECT: Design and Analysis of a ZVS Totem-Pole PFC Integrated Resonant Converter for Low-Voltage Battery Charging Applications.

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Design and Analysis of a ZVS Totem-Pole PFC Integrated Resonant Converter for Low-Voltage Battery Charging Applications

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Abstract—The growing demand for compact, energy-efficient, and high-performance low-voltage DC power supplies in modern applications such as consumer electronics, industrial automation, and distributed power systems necessitates the development of advanced power conversion topologies. This paper presents the design and analysis of a high-efficiency two-stage AC-DC converter consisting of a Zero-Voltage Switching (ZVS) based Totem-Pole power factor correction (PFC) stage followed by a half-bridge LLC resonant converter. The PFC stage maintains a voltage of 150 V which is regulated to charge a 25 V device by the LLC resonant converter. The output of the proposed system is regulated at 25 V with a load current of 5.8 A. The totem-pole PFC operates in continuous conduction mode (CCM) to achieve near-unity power factor and stable DC-link voltage. In comparison to conventional boost PFCs, the bridgeless totem-pole eliminates the diode bridge, thereby reducing conduction losses and improving efficiency. The second stage employs a half-bridge LLC resonant converter to provide galvanic isolation and a regulated low-voltage output. The half-bridge topology reduces the number of required switches while maintaining high conversion efficiency. The proposed system is modeled and designed using MATLAB/Simulink, and its performance is evaluated under steady-state and dynamic conditions. The proposed model offers a compact and reliable solution for modern low-voltage DC power applications.

Index Terms—Keywords— Totem-Pole PFC, Zero Voltage Switching (ZVS), LLC Resonant Converter, Low-Voltage Power Supply, Power Factor Correction.

I. INTRODUCTION

Modern electrical systems increasingly demand high efficiency, compact size, and superior power quality; however, conventional power conversion techniques often suffer from significant conduction and switching losses, high total harmonic distortion (THD), and reduced performance at high switching frequencies. Over the years, various converter topologies have been developed to meet diverse operational requirements. At present, low-cost battery charging systems often rely on conventional diode bridge rectifier (DBR)-based front-end architectures, which introduce significant input current distortion and poor power factor [1]. Conventional converter topologies, including diode bridge rectifiers followed by boost converters and isolated DC-DC converters, offer simplicity but suffer from higher conduction losses, increased component count, and limited efficiency at high frequencies. As a result, two-stage conversion architectures have gained

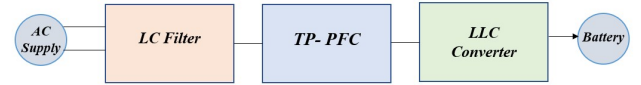


Fig. 1. Block diagram of two-stage battery charger

prominence, where the first stage performs power factor correction and the second stage provides regulated DC output with galvanic isolation [2].

Among various DC-DC converter configurations, the LLC resonant converter has emerged as a preferred choice due to its high efficiency and ability to operate at high switching frequencies. This is primarily enabled by soft-switching techniques such as zero-voltage switching (ZVS), which significantly reduce switching losses. Additionally, high-frequency operation allows the use of smaller passive components, resulting in compact and high-power-density designs suitable for applications ranging from consumer electronics to electric vehicle charging systems [3].

Modern electronic systems, ranging from compact charging adapters to high-power battery charging units, draw energy from AC sources and often behave as nonlinear loads. As a result, in the absence of power factor correction (PFC), these systems inject harmonic-rich currents into the grid, degrading power quality and imposing additional stress on distribution transformers and other utility equipment. To mitigate these effects and ensure compliance with power quality standards, a PFC stage is integrated at the front end of the system, as demonstrated in Fig. 1. Among front-end PFC options, the totem-pole PFC offers several advantages over conventional boost PFC. Its bridgeless topology eliminates the input diode bridge, reducing conduction losses and improving efficiency. It supports zero-voltage switching (ZVS), minimizing switching losses and enabling high-frequency operation [4]. Totempole based PFC is generally used for high power designs. Traditional Totempole based PFC circuits has complex control system. This type of chargers often faces medium voltage and current ripple. Totempole based PFC is generally used for high power. To address this issue we proceed the work with Interleaved ZVS Totempole PFC based charger for low

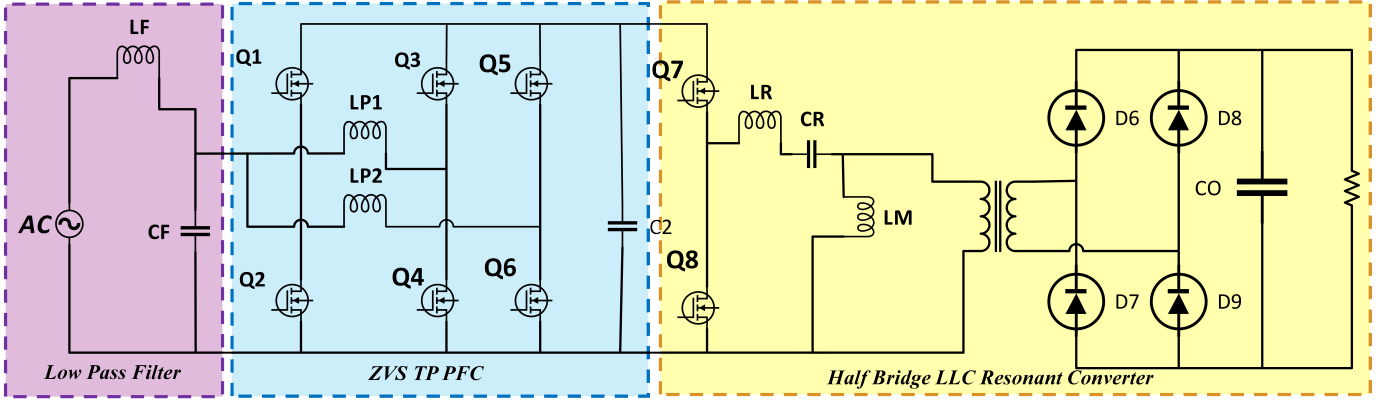


Fig. 2. Simulation model of the proposed Totem-Pole PFC assisted half-bridge LLC resonant converter.

power application. On the DC–DC conversion side, the LLC resonant converter has emerged as the preferred choice due to its high efficiency and ability to operate at high switching frequencies making it suitable for applications ranging from laptop adapters to electric vehicle chargers [5], [6].

Therefore, this paper addresses the design gap for compact, low-voltage battery charging applications by proposing a fully integrated ZVS totem-pole PFC and half-bridge LLC resonant converter system for regulated DC output and isolation. The objective of this paper is to design and analyze a high-efficiency two-stage power conversion system that achieves improved power quality, reduced losses, and high power density through the integration of ZVS-based totem-pole PFC and LLC resonant converter topologies.

II. SYSTEM CONFIGURATION

The proposed battery charger configuration consists of an input LC filter, a zero-voltage-switching (ZVS) totem-pole power factor correction (PFC) converter at the primary stage, and a half-bridge LLC resonant converter at the secondary stage as depicted in Fig. 2. The front-end LC filter is employed to suppress high-frequency switching harmonics and mitigate electromagnetic interference (EMI) reflected back to the supply. The filtered AC input is processed by the totem-pole PFC converter, which utilizes a bridgeless structure with high-frequency switches operated in a complementary manner with an effective 50% duty cycle, ensuring symmetrical operation and reduced input current ripple. This mode of operation contributes to improved current shaping and high efficiency while maintaining near-unity power factor through a closed-loop control scheme incorporating voltage and current feedback [7].

The regulated DC-link output feeds the half-bridge LLC resonant converter, where two switches operate in complementary fashion to excite the resonant tank composed of resonant inductance, magnetizing inductance, and resonant capacitance. The switching frequency is selected relative to the resonant frequency to ensure ZVS operation, thereby minimizing switching losses and enhancing overall efficiency [8]. The high-frequency transformer provides galvanic isolation and

voltage scaling, while the secondary side employs rectification and an output capacitor filter to deliver a stable DC output. The overall system is designed to achieve reduced switching stress, minimized input current ripple, and high power density through effective integration of the LC filter, totem-pole PFC stage, and LLC resonant converter.

III. DESIGN METHODOLOGY AND THEORETICAL FRAMEWORK

This section details the design and implementation of a Totem-Pole power factor correction (PFC) assisted half-bridge LLC resonant converter for battery charging applications. The proposed methodology emphasizes high efficiency, zero-voltage switching (ZVS) operation, and compact, high-power-density design. For simplification and accurate parameter determination, the design assumes ideal semiconductor devices and models the load as the battery's equivalent resistance. The integration of the Totem-Pole PFC stage with the LLC resonant converter ensures improved power quality, reduced conduction and switching losses, and reliable performance under varying load conditions, forming the basis for the overall high-efficiency charging system.

A. Totempole PFC

The design parameter of totem-pole PFC is calculated by considering the following assumption, that the recommended topology contains ideal semiconductor devices. Also, for the input inductor, the Totem-Pole PFC is intended to function in continuous conduction mode (CCM) [9].

For a totem-pole power factor correction (PFC) converter with voltage ripple $\%Ripple_V = 1\%$ and current ripple $\%Ripple_I = 4\%$, the boost inductance and the maximum inductor current are determined as follows.

The boost inductance L (in henries) is given by

$$L = \frac{1}{\%Ripple_V} \cdot \frac{V_{ac}^2}{P_0} \left(1 - \frac{\sqrt{2} \cdot V_{ac}}{V_0} \right) \cdot T, \quad (1)$$

where V_{ac} is the RMS input AC voltage (in volts), P_0 is the output power (in watts), V_0 is the DC output voltage (in volts), and T is the switching period (in seconds).

The maximum peak inductor current $I_{L,\max}$ (in amperes) is expressed as

$$I_{L,\max} = \frac{\sqrt{2} \cdot P_0}{V_{ac}} \cdot \left(1 + \frac{\%Ripple_I}{2}\right), \quad (2)$$

where the current ripple is set to $\%Ripple_I = 4\%$.

B. LLC

The LLC resonant converter design requires several key parameters, including the equivalent load, quality factor, resonant frequency, and transformer design parameters [10]. The equations used are given below.

The equivalent load seen at the primary of the transformer, R_{ac} , is calculated as:

$$R_{ac} = \frac{8}{\pi^2} \cdot \frac{N_p^2}{N_s^2} \cdot \frac{V_{out}^2}{P} \quad (3)$$

Here, N_p is the number of primary turns of the transformer, N_s is the number of secondary turns, V_{out} is the output voltage, and P is the output power of the converter.

The quality factor Q of the resonant tank is given by:

$$Q = \frac{1}{R_{ac}} \sqrt{\frac{L_r}{C_r}} \quad (4)$$

Here, L_r is the resonant inductor, C_r is the resonant capacitor, and R_{ac} is the equivalent load as defined in Eq. (4).

The resonant frequency of the LLC converter is:

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (5)$$

where f_r is in Hertz.

The transformer magnetizing ratio m is calculated as:

$$m = \frac{L_m + L_r}{L_r} \quad (6)$$

where L_m is the magnetizing inductance of the transformer. The higher the m , the lesser the circulating current in the converter.

The transformer turns ratio n is:

$$n = \frac{V_{dc_link}}{2V_{out}} \quad (7)$$

where V_{dc_link} is the input DC link voltage. The center tap diode rectifier is used instead of a full diode bridge rectifier to reduce the voltage drop and size of the converter on the secondary side. ZVS is obtained when the switching frequency is operated near the resonant frequency.

The ratio of switching frequency to resonant frequency

$$F = \frac{f_s}{f_r} \quad (8)$$

where f_s is the switching frequency.

The maximum voltage gain occurs at:

$$f_p = \frac{1}{2\pi\sqrt{(L_m + L_r) C_r}} \quad (9)$$

where f_p is the parallel resonant frequency.

C. LC Filter Design Equations

The LC filter is used to reduce the ripple in the output current and voltage. The design equations are as follows:

The ideal filter capacitance C_f is given by:

$$C_f = \frac{I_{max}}{2f_{line} V_{max} \tan \theta} \quad (10)$$

Here, I_{max} is the peak supply current, V_{max} is the peak supply voltage, f_{line} is the supply frequency (typically 50 Hz), and θ is the filter phase angle.

The filter inductance L_f is calculated as:

$$L_f = \frac{1}{4\pi^2 f_c^2 C_f} \quad (11)$$

Here, f_c is the corner frequency of the filter, and C_f is the filter capacitance as defined in Eq. (11).

IV. SIMULATION AND RESULT ANALYSIS

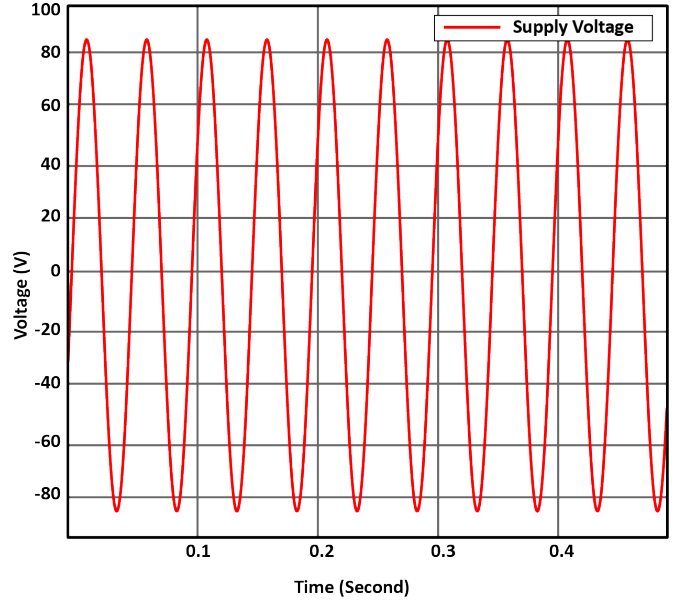


Fig. 3. Supply Voltage.

The converter is modeled in MATLAB/SIMULINK to validate the functioning of the converter. The specifications of the converter and designed parameters are specified in Table I. The 110 Vrms supply is converted to 150V DC link voltage by the ZVS TP PFC. The duty ratio is chosen as 0.5 to meet the condition. The switching frequency of the primary stage and secondary stage converter is chosen as 80kHz. The resonant frequency of the LLC resonant converter is 45kHz. The optimal voltage for charging the battery can be derived when operated at the frequency range of 46kHz to 50kHz. The inclusion of input filter component further ensures reduced ripple and improved power quality. Specifically, the output current ripple is limited to 4%, demonstrating effective filtering and stable output performance. The transformer turns ratio of 3.39 is selected to step down the DC-link voltage to the

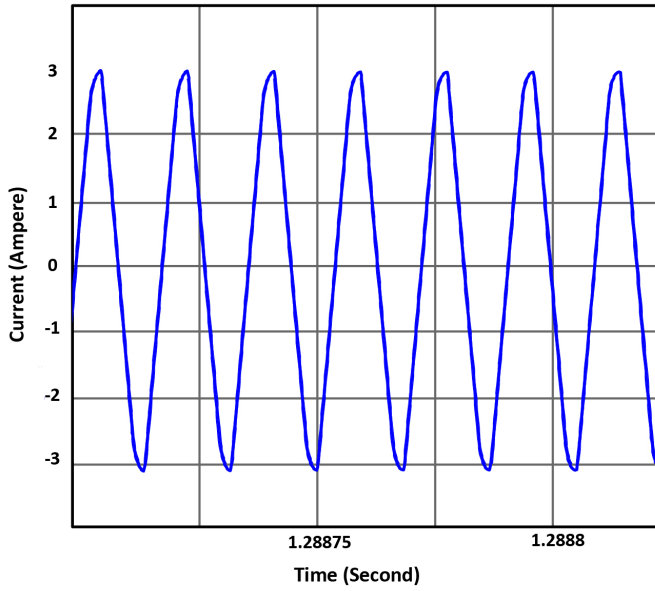


Fig. 4. Magnetizing Current of LLC

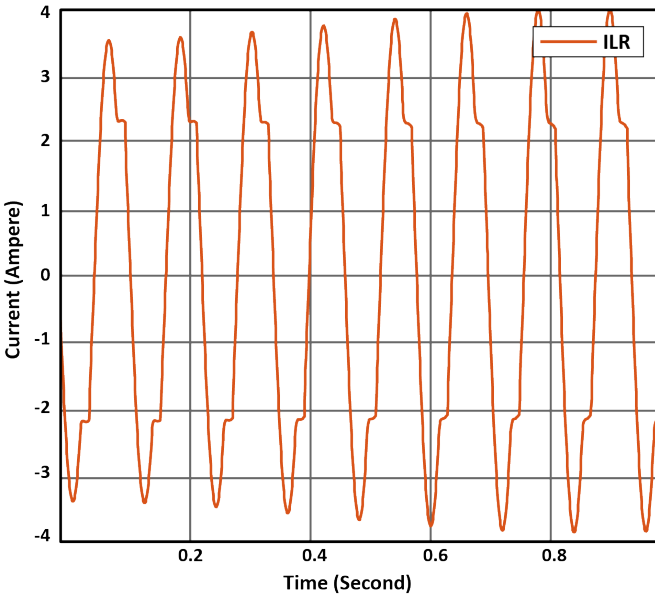


Fig. 5. Resonant Current of LLC

required low-voltage level while maintaining efficient energy transfer.

The magnetizing and resonant inductor current waveforms shown in Fig. 4 and Fig. 5 respectively exhibit high-frequency oscillatory behavior, consistent with LLC resonant tank operation. The magnetizing current shows a relatively smooth, lower-amplitude waveform, indicating proper energy storage and ZVS support. The resonant inductor current has a higher amplitude and sinusoidal nature, confirming effective resonance between tank components.

The DC-link voltage waveform of the interleaved totem-

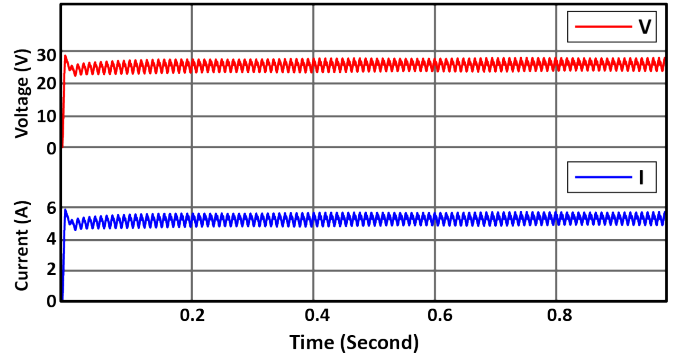


Fig. 6. Output Voltage and Current of battery charger

pole PFC shows a well-regulated output centered around the desired reference level. The voltage exhibits small high-frequency ripple components due to switching action, which are effectively minimized by interleaving. The reduced ripple amplitude indicates improved current sharing between phases and enhanced filtering performance. Overall, the waveform confirms stable DC-link regulation with low ripple and high-quality output voltage. The output voltage waveform shows a well-regulated DC level at approximately 25 V with minimal ripple, indicating effective voltage regulation by the LLC converter as depicted in Fig. 6. The output current waveform remains steady around 5.8 A, confirming stable load conditions and proper current delivery. Both voltage and current exhibit low ripple, demonstrating efficient filtering and stable steady-state operation. These results validate the performance of the charger in providing a stable and reliable DC output.

Table II presents the key current and power characteristics of the proposed converter, demonstrating its capability to deliver stable and efficient performance. The system maintains a regulated DC-link voltage of 150 V, ensuring a consistent input to the downstream LLC resonant stage.

The converter achieves an output voltage of 25 V with a corresponding load current of 5.8 A, resulting in an output power of approximately 1 kW. These values confirm that the system meets the intended low-voltage, high-current application requirements. The RMS current values (15.82 A and 17.32 A) indicate proper current handling capability under operating conditions, while the average current values reflect steady-state operation with minimal fluctuations.

V. CONCLUSION

This paper presented the design and analysis of a high-efficiency two-stage AC–DC power conversion system tailored for low-voltage DC applications. The proposed architecture integrates a Zero-Voltage Switching (ZVS)-based totem-pole power factor correction (PFC) stage with a half-bridge LLC resonant converter, enabling improved efficiency, reduced losses, and compact system design. The totem-pole PFC stage, operating in continuous conduction mode (CCM), effectively maintains a stable DC-link voltage while achieving near-unity power factor. By eliminating the conventional diode bridge,

TABLE I
SPECIFICATIONS AND COMPONENTS OF CONVERTER

Selected Parameters	Symbol	Value
Input Voltage	V_{inp}	110 Vrms
Output Current Ripple	ΔI_{out}	4%
Designed Parameters	Symbol	Value
Duty Cycle	D	0.5
TPPFC Inductance	L_{p1} and L_{p2}	3.04 μ H
DC Link Capacitor	C_{p1}	880 μ F
Output Filter Capacitor	C_{s1}	40 μ F
Resonant Inductance	L_r	74 μ H
Magnetizing Inductance	L_m	222 μ H
Resonant Capacitor	C_r	169nF
Input Low Pass Filter Inductance	L_f	330 μ H
Input Low Pass Filter Capacitor	C_f	3 μ F
Switching Frequency	f_s	80kHz
Resonant Frequency	f_r	45kHz
Transformer Turns Ratio	n	3.39

TABLE II
CURRENT AND POWER SPECIFICATIONS

Parameter	Value
RMS Current	15.82, 17.32 A
Average Current	1.0204, 1.0097 A
DC link voltage	150 V
Output Power	1 kW
Output Voltage	25 V
Output Current	5.8 A

the bridgeless configuration significantly reduces conduction losses and enhances overall system performance.

The subsequent half-bridge LLC resonant converter provides galvanic isolation and regulates the DC-link voltage to a low-voltage output, ensuring efficient power delivery to the load. Its inherent soft-switching capability and reduced switch count contribute to high conversion efficiency and simplified implementation. The system is designed to regulate the output at 25 V with a load current of 5.8 A, demonstrating its suitability for practical low-voltage applications.

Simulation results obtained from MATLAB/Simulink validate the effectiveness of the proposed design under both steady-state and dynamic operating conditions. Overall, the proposed converter offers a reliable, efficient, and compact solution for modern low-voltage DC power supply systems, making it a promising candidate for applications in consumer electronics, industrial systems, and distributed power architectures.

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Project Development Stages

1. Requirement Specification

Define system objectives: AC–DC conversion for low-voltage battery charging with 150 V DC-link and 25 V, 5.8 A output, high efficiency, compact size, and near-unity power factor.

2. Literature Review and Problem Analysis

Analyze conventional boost PFC and isolated DC-DC converters. Identify limitations such as higher conduction losses and lower efficiency, motivating the use of bridgeless TP-PFC and LLC topology.

3. Topology Selection and System Design

Select a two-stage architecture:

Stage 1: ZVS Totem-Pole PFC (CCM operation)

Stage 2: Half-Bridge LLC Resonant Converter

Developed overall block diagram and define operating principles.

4. Design Calculations and Component Selection

Design key parameters:

PFC inductor, switching devices, DC-link capacitor

LLC resonant tank (L_r , L_m , C_r), transformer turns ratio voltage gain.

5. Modeling and Simulation (MATLAB/Simulink)

Develop complete simulation model including control circuits.

Simulate PFC stage (power factor correction, DC-link regulation) and LLC stage (resonant behavior, output regulation).

6. Performance Evaluation

Analyze system under Steady-state operation

Evaluate waveforms: inductor current, DC-link voltage, resonant currents, output voltage/current.

7. Results Validation

Confirm: Output regulation at 25 V, 5.8 A

Stable 150 V DC-link voltage and reduced ripple

8. Conclusion and Future Work

Summarize improvements in efficiency, compactness, and reliability.

Suggest future scope such as hardware implementation, digital control, and optimization techniques.

Task Allocation Among Team Members

Member Name(ID)	Task
Ashfaqur Rahman(202216012)	Designing Circuit, Modeling, Simulation
Jahin Shahriar(202216021)	Designing Circuit, Simulation
Md. Raisul Islam Ratul(202216049)	Paper writing, Presentation
Mubashwira Nawar(202216066)	Modeling, Paper writing, Literature Review,

Tools and Software Used

1. Simulation and Modeling

- MATLAB/Simulink
 - Used for complete system modeling of TP-PFC and LLC converter
 - Simulation of steady-state and dynamic performance
 - Waveform analysis (inductor current, DC-link voltage, output)
- Simscape Electrical
 - Used for accurate power electronics component modeling
 - Implementation of switches, inductors, transformers, and control blocks

2. Design and Analysis Tools

- MATLAB
 - Used for design calculations (inductor, capacitor, resonant tank)
 - Plotting gain curves and analyzing system parameters

3. Documentation and Writing

- Microsoft Word / LaTeX (Overleaf)
 - Used for writing the project report and conference paper
 - Formatting according to IEEE standards

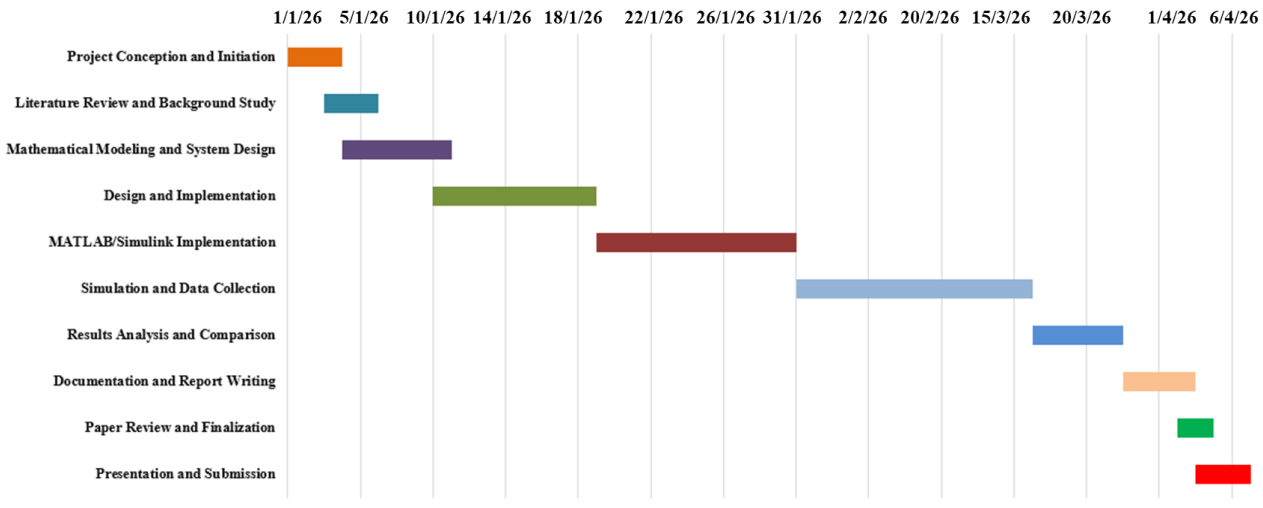
4. Presentation, Diagram and Visualization Tools

- Microsoft PowerPoint .Microsoft Project
 - Used for block diagrams, circuit diagrams, Gantt Chart and presentation slides

PROJECT TIMELINE REPORT

The project is structured to transition from initial theoretical research into technical implementation, followed by extensive simulation and final documentation. The timeline has been optimized to ensure that foundational tasks are completed early, allowing for a robust testing and analysis phase.

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Phase Breakdown

Phase I: Research and Design

The project commences with Project Conception and Initiation. A Literature Review is conducted concurrently with Mathematical Modeling and System Design. This overlapping structure ensures that the system design is directly informed by current research, establishing a solid theoretical framework within the first two weeks.

Phase II: Development and Implementation

Upon finalization of the mathematical models, the project moves into Controller Design and Implementation. This is followed by a 19-day period dedicated to MATLAB/Simulink Implementation, where the theoretical logic is translated into a functional simulation environment.

Phase III: Testing and Analysis

The core empirical work is centered on Simulation and Data Collection, which is allocated the largest single block of time (23 days). Following data acquisition, a period of Results Analysis and Comparison is scheduled to validate the findings against project objectives and existing benchmarks.

Phase IV: Finalization and Submission

The concluding phase focuses on the formalization of results. Documentation and Report Writing is followed by a Paper Review and Finalization stage. The project concludes with the final Presentation and Submission on April 6, 2026.

Future Work

Future work can extend the proposed ZVS Totem-Pole PFC integrated LLC converter in several directions to enhance performance, scalability, and practical applicability:

Advanced Converter Topologies:

Future research may explore multi-level and interleaved PFC structures to reduce voltage stress on switches and improve current sharing. Additionally, advanced resonant topologies such as full-bridge LLC or hybrid resonant converters can be investigated to achieve wider gain range and improved regulation under varying load conditions.

Higher Efficiency Designs:

Further optimization can be achieved through soft-switching enhancement, reduction of circulating currents in the resonant tank, and improved magnetic design. Techniques such as adaptive dead-time control, synchronous rectification, and optimized modulation strategies can significantly reduce switching and conduction losses.

Integration with Renewable Energy Systems:

The proposed converter can be extended for integration with renewable sources such as photovoltaic (PV) systems and battery energy storage systems. This includes implementing maximum power point tracking (MPPT), bidirectional power flow capability, and compatibility with DC microgrid architectures.

Use of Advanced Semiconductor Devices:

Replacing conventional silicon devices with wide bandgap (WBG) semiconductors such as SiC and GaN can enable higher switching frequencies, reduced switching losses, and improved thermal performance. This would also allow further reduction in passive component size, leading to higher power density.

Expansion to Higher Power Applications:

The design can be scaled for higher power levels by adopting modular or parallel converter structures. Special attention can be given to thermal management, electromagnetic interference (EMI) mitigation, and robust control strategies to ensure stable operation in high-power industrial or electric vehicle (EV) charging applications.

Digital Control Implementation:

Future work may include implementation of digital controllers using DSP or FPGA platforms for precise control, real-time monitoring, and adaptive tuning of the converter under varying operating conditions.

Hardware Prototyping and Experimental Validation:

A practical hardware prototype can be developed to validate simulation results. Experimental analysis of efficiency, power factor, thermal behavior, and dynamic response will strengthen the reliability of the proposed design.